Sensorless Control of PMSM via ADRC and SMC with Super-Twisting Observer

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***Abstract*—This study explores in depth an innovative control strategy specifically for permanent magnet synchronous motors (PMSMs) that do not require position sensors. The control strategy combines advanced technologies of superspiral sliding mode observer (STSMO), active anti-disturbance control (ADRC) and sliding mode controller (SMC). Among them, STSMO, with its excellent observation algorithm, can provide accurate estimates of motor state in complex environments full of high-frequency noise and constantly changing system parameters. On this basis, we design a sliding mode controller to adjust the motor current, which ensures fast response speed and high control accuracy. At the same time, the introduction of ADRC further optimizes motor speed regulation, significantly improving the robustness of the system by estimating and compensating for internal motor uncertainties and external disturbances. Through simulation experiments, we verify the effectiveness of this control strategy, and confirm that it can achieve high performance control of PMSMs under sensorless conditions, showing excellent dynamic response speed and excellent steady-state performance.**

Keywords-Permanent Magnet Synchronous Motor (PMSM), Super-Twisting Sliding Mode Observer (STSMO), Active Disturbance Rejection Control (ADRC), Sliding Mode Controller (SMC)

I. SUPER-TWISTING SLIDING MODE OBSERVER DESIGN

In various fields such as industrial automation, electric vehicles, and wind power generation, Permanent Magnet Synchronous Motors (PMSM) are widely used due to their high efficiency, high power density, and excellent dynamic performance[1]. However, traditional PMSM control systems rely on position sensors to obtain rotor position information, which not only increases costs but also the complexity of sensor installation[2]. Therefore, researchers have begun to turn to sensorless control methods in order to reduce costs and simplify system design. This paper will focus on sensorless control strategies based on motor back electromotive force, especially in the medium and high-speed control range.

In the field of medium and high-speed control, it is crucial to accurately estimate the rotor position and speed of the motor, especially in the absence of a position sensor. The Sliding Mode Observer (SMO) is widely used for such estimation tasks due to its strong robustness and fast response[3]. However, traditional SMO has a chattering problem, which may affect the stability and control accuracy of the system. To address this issue, this paper uses an improved observer—the Super-Twisting Sliding Mode Observer (STSMO). The STSMO suppresses chattering and reduces the phase delay of speed and position estimation by designing a sliding surface and a super-twisting control law, utilizing its second-order sliding mode characteristics, thereby enhancing the system's robustness and dynamic response speed[4].

Despite the progress made in the design of STSMO, relying solely on it for control is not sufficient to address all the challenges in the PMSM system. The parameters of the PMSM system's motor are often affected by working conditions and load changes, and the traditional PI control strategy struggles to adapt to this dynamically changing environment, making it difficult to meet the requirements for high precision and strong robustness. To overcome these limitations, this paper introduces two advanced control methods: Active Disturbance Rejection Control (ADRC) and Sliding Mode Control (SMC). The core of ADRC lies in estimating and compensating for the total disturbance of the system in real time through the Extended State Observer (ESO)[5], thereby achieving precise control of the PMSM system with strong nonlinearity and uncertainty without relying on an accurate mathematical model[6]. SMC, as a nonlinear control strategy, makes the system state slide along the sliding surface by designing a sliding surface and switching control law, but its inherent chattering phenomenon may affect the stability and control accuracy of the system[7]. To address the chattering issue in SMC, this paper combines the control strategy of STSMO to achieve accurate observation of the system state and reduce the impact of chattering. This combined method not only maintains the robustness and rapid response characteristics of SMC but also improves the system's stability and control accuracy.

This paper integrates the control strategies of ADRC, SMC, and STSMO to better handle the nonlinearity, parameter uncertainty, and external disturbances in the PMSM system. Through this integrated approach, this paper aims to analyze the potential and challenges of these control strategies in practical engineering applications, providing a new solution for sensorless control of PMSM.

II. SUPER-TWISTING SLIDING MODE OBSERVER DESIGN

In this paper, an improved type of sliding mode observer, known as the Super-Twisting Sliding Mode Observer (STSMO), is employed. It utilizes a second-order sliding mode control strategy, effectively suppressing system chattering by

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incorporating an integral term and a nonlinear term, while ensuring the system's rapid response capability. The design of this observer is crucial for the control of high-speed Permanent Magnet Synchronous Motors (PMSM), especially in sensorless applications. The voltage equations of the high-speed Permanent Magnet Synchronous Motor (PMSM) in the d,q-axis coordinate system are as follows:

$$\begin{bmatrix} u_d \\ u_q \end{bmatrix} = \begin{bmatrix} R_s & -\omega_e L_q \\ \omega_e L_q & R_s \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + p \begin{bmatrix} \psi_d \\ \psi_q \end{bmatrix} + \begin{bmatrix} E_\alpha \\ E_\beta \end{bmatrix} \tag{1}$$

In the equation: u_d and u_q represent the stator voltages in the d,q -axis coordinate system; i_d and i_q represent the stator currents in the d,q -axis coordinate system. is the differential operator, ψ_d and ψ_q are the stator flux linkages in the d,q -axis coordinate system; L_d and L_q are the stator inductances in the d,q -axis coordinate system; ω_e is the motor's angular velocity, E_α and E_β are the extended back electromotive forces (EMFs), and they satisfy the following conditions:

$$\begin{bmatrix} E_\alpha \\ E_\beta \end{bmatrix} = \left[(L_d - L_q)(\omega_e i_d - p i_q) + \omega_e \psi_f \right] \begin{bmatrix} -\sin\theta_e \\ \cos\theta_e \end{bmatrix} \tag{2}$$

From equation 2, it can be seen that the extended back electromotive force (EMF) of the PMSM contains all the information about the motor's rotor position and speed. Therefore, only by accurately obtaining the extended back EMF can the full information of the motor speed be calculated[8].

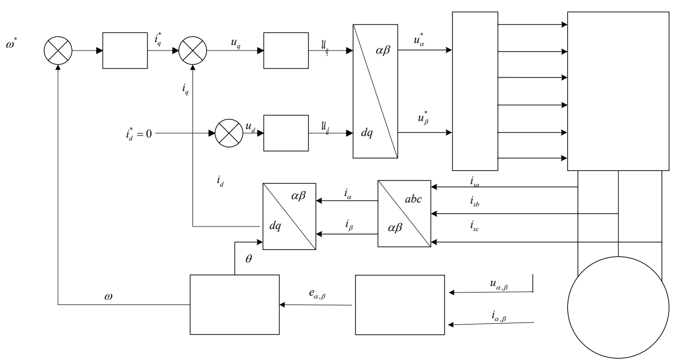

Figure 1: PMSM Vector Control Block Diagram

The stator currents i_d and i_q of the Permanent Magnet Synchronous Motor (PMSM) are coupled in the d-axis and d-axis directions, forming the back electromotive force (EMF). In order to achieve better control characteristics for the PMSM, This paper adopts the vector control strategy of $i_d = 0$. The control block diagram is shown in Figure 1.

This paper adopts the difference between the estimated currents and the measured currents in the stationary coordinate system as the current observation error $\tilde{i}_\alpha$ and $\tilde{i}_\beta$, and defines it as the sliding surface[9].

$$\begin{cases} \tilde{i}_\alpha = i_\alpha^* - i_\alpha \\ \tilde{i}_\beta = i_\beta^* - i_\beta \end{cases} \tag{3}$$

Where i_β and i_α are the measured currents, i_α^* and i_β^* are the estimated currents. Once the observer's state variables reach the sliding surface, they will remain on it. According to the control law of the sliding surface, the estimated value of the back electromotive force can be expressed as:

$$\begin{cases} \hat{e}_\alpha = \dfrac{k_1}{L_d} \left|\tilde{i}_\alpha\right|^{\frac{1}{2}} sign(\tilde{i}_\alpha) + \dfrac{1}{L_d} \displaystyle\int k_2 sign(\tilde{i}_\alpha) dt \\ \hat{e}_\beta = \dfrac{k_1}{L_d} \left|\tilde{i}_\beta\right|^{\frac{1}{2}} sign(\tilde{i}_\beta) + \dfrac{1}{L_d} \displaystyle\int k_2 sign(\tilde{i}_\beta) dt \end{cases} \tag{4}$$

In the equation: $\hat{e}_\alpha$ and $\hat{e}_\beta$ are the estimated values of the α and β axis back electromotive forces, respectively.

The STSMO illustrates how the observer achieves accurate observation of the system state through a second-order sliding mode control strategy and effectively suppresses chattering. With this design, the STSMO presented in this paper not only enhances the system's robustness and response speed but also provides an effective solution for sensorless control of high-speed PMSMs.

Although the STSMO can suppress chattering, sliding mode control can still generate high-frequency oscillations that affect the estimation of the back electromotive force, thereby amplifying the angle estimation error. Therefore, this section employs a Phase-locked Loop (PLL) system to extract the rotor position information and reduce the angle estimation error.

The back electromotive force (EMF) signals obtained through the STSMO can be used to calculate the motor's angle and position. First, the back EMF signals $\hat{e}_\alpha$ and $\hat{e}_\beta$ are extracted. Then, the arctangent function is utilized to calculate the motor's angle $\hat{\theta}_e$ and angular velocity $\hat{\omega}_e$, as shown in Equation 5.

$$\begin{cases} \hat{\theta}_e = -\arctan(\dfrac{\hat{e}_\beta}{\hat{e}_\alpha}) \\ \hat{\omega}_e = \dfrac{d\hat{\theta}_e}{dt} \end{cases} \tag{5}$$

Since the output range of the arctangent function is between $(-\frac{\pi}{2}, \frac{\pi}{2})$, and the actual motor position range is between $(0, 2\pi)$. Therefore, a compensation value needs to be added or subtracted based on the sign of $\hat{e}_\beta$, transforming the original formula into Equation 6.

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$$\begin{cases} \hat{\theta}_e = -\arctan(\frac{\hat{e}_\beta}{\hat{e}_\alpha}) + \frac{-\hat{e}_\beta + |\hat{e}_\beta|}{2|\hat{e}_\beta|} \cdot \pi \\ \hat{\omega}_e = \frac{d\hat{\theta}_e}{dt} \end{cases} \tag{6}$$

By using the aforementioned method, it is possible to accurately estimate the motor's angle and position, thereby further improving the system's control precision and stability.

III. Design of ADRC and SMC Control Strategies

A. Design of the Speed Loop ADRC

ADRC is a control method that does not depend on an accurate mathematical model, composed of a Tracking Differentiator (TD), an Extended State Observer (ESO), and a Nonlinear State Error Feedback (NLSEF) control law. The core concept of ADRC is to regard the uncertainty and external disturbances in the system as the "total disturbance", and to estimate and compensate for this disturbance in real time through the Extended State Observer (ESO), thus achieving robust control of the system[10]..

1) Design of the Tracking Differentiator (TD)

The function of the Tracking Differentiator (TD) is to arrange the given reference signal with a reasonable transition, making the input signal to the controller smoother. The mathematical equation for the speed of the PMSM is given by Equation 7:

$$\dot{\omega} = \frac{3p\psi_f i_q}{2J} - \frac{B}{J}\omega - \frac{T_L}{J} \tag{7}$$

In the equation: ψ_f represents the magnetic flux linkage of the permanent magnet; J represents the moment of inertia; B represents the damping coefficient; T_L represents the load torque.

Let $f(\omega) = -\frac{B}{J}\omega$, $b_0 = \frac{3p\psi_f i_q}{2J}$, $\gamma_s(t) = -\frac{T_L}{J}$, and consider $\gamma_s(t)$ as the system disturbance. To match the Active Disturbance Rejection Controller (ADRC) with the speed model, a nonlinear function can be used to replace $\gamma_s(t)$. The resulting Equation 8 is as follows:

$$\dot{\omega} = f(\omega) + \gamma_s(t) + b_0 i_q \tag{8}$$

Then the Tracking Differentiator (TD) of the ADRC is as shown in Equation 9:

$$\begin{cases} e_{s0} = \omega^* - v_{s1} \\ \dot{v}_{s1} = -\gamma_{s0} fal(e_{s0}, \alpha_{s0}, \delta_{s0}) \end{cases} \tag{9}$$

In the equation: e_{s0} represents the error between the target speed value $\dot{v}_{s1}$ and the actual speed value in the speed loop, which is input into the nonlinear function. By appropriately adjusting the gain coefficient, it tracks the speed signal; γ_{s0} is the nonlinear adjustment gain value of the ADRC's Tracking Differentiator signal function.

2) Design of the Extended State Observer (ESO)

ESO is an observer designed to estimate the system's state. By combining the system model with measured data, it estimates the unknown state variables and unmeasured disturbances in real time, thereby providing an accurate estimation of the system's state for use in control systems. The Extended State Observer (ESO) is given by Equation 10:

$$\begin{cases} e_{s1} = z_{s1} - \omega \\ \dot{z}_{s1} = z_{s2} - \beta_{s1} fal(e_{s1}, \alpha_{s1}, \delta_{s1}) + f_s(z_{s1}) + b_0 i_q^* \\ \dot{z}_{s2} = -\beta_{s2} fal(e_{s1}, \alpha_{s1}, \delta_{s1}) \end{cases} \tag{10}$$

In the equation: e_{s1} represents the error between the estimated speed of the Extended State Observer (ESO) and the actual feedback speed. This error is used as the input signal for the nonlinear function in the ESO. β_{s1} and β_{s2} represent the gain coefficients for the tracking signal's derivative and the disturbance error estimation, respectively.

3) Nonlinear State Error Feedback Control Law (NLSEF)

In NLSEF, the key lies in nonlinear processing of error signals to achieve more precise control. The core idea of NLSEF is to use nonlinear function to feedback and compensate the error. The control law equation is shown in equation 11:

$$\begin{cases} e_{s2} = v_{s1} - z_{s1} \\ i_{q0}(t) = k_s fal(e_{s2}, \alpha_{s2}, \beta_{s3}) \\ i_q^*(t) = i_{q0}(t) - z_{s2}/b_0 \end{cases} \tag{11}$$

In the equation: k_s is the gain coefficient of the control law; e_{s2} represents the error between the tracking signal of the Tracking Differentiator and the estimated signal of the Extended State Observer, with this error serving as the input to the control law; b_0 is the factor used in the Extended State Observer to compensate for the disturbance estimation.

B. Design of the Current Loop Sliding Mode Control

1) Design of Current Loop Feedforward Decoupling

Decoupling design in the current loop of Permanent Magnet Synchronous Motor (PMSM) control is of great significance. Through decoupling, the cross-coupling effects between the d-axis and q-axis can be eliminated, making the motor control more flexible and precise. After decoupling, the current loop can control the d-axis current and q-axis current separately, allowing the system to respond better under different working conditions, thereby improving the system's performance and stability.

To obtain the expression of the current controller, it is

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necessary to write out the current equations in the i_d and i_q coordinate systems as shown in Equation 12:

$$\begin{cases} \dfrac{d}{dt} i_d = -\dfrac{R}{L_d} i_d + \dfrac{L_q}{L_d} \omega_e i_q + \dfrac{1}{L_d} u_d \\ \dfrac{d}{dt} i_q = -\dfrac{R}{L_d} i_d - \dfrac{1}{L_d} \omega_e \left(L_d i_d \psi_f \right) + \dfrac{1}{L_q} u_q \end{cases} \tag{12}$$

From the above equation, it can be seen that the stator currents i_d and i_q have cross-coupling relationships in the d-axis and q-axis directions. This cross-coupling can lead to inflexible system response and low control accuracy. By decoupling to eliminate the coupling between the d-axis and q-axis, the independence and precision of motor control are enhanced, thereby improving performance, stability, response speed, and control accuracy. This paper adopts a conventional decoupling control strategy, and the voltages for the d-axis and q-axis are as follows:

$$\begin{cases} u_d^* = \left(K_{Pd} + \dfrac{K_{Id}}{s} \right) \left(i_d^* - i_d \right) - \omega_e L_q i_q \\ u_q^* = \left(K_{Pq} + \dfrac{K_{Iq}}{s} \right) \left(i_q^* - i_q \right) + \omega_e \left(L_d i_d + \psi_f \right) \end{cases} \tag{13}$$

Where: K_{Id} and K_{Iq} are the proportional gains of the PI controllers, K_{Pd} and K_{Pq} are the integral gains of the PI controllers.

2) Current Loop Sliding Mode Control (SMC) Design

Designing the sliding mode surface and the switching control law through Sliding Mode Control (SMC) is a key step in improving the control performance of the PMSM current loop. The basic principle is to first construct a sliding mode surface, which is composed of the error between the system state and the desired value. Then, a switching control law is designed to make the system state move along the sliding mode surface. The defined current sliding mode surface function can effectively guide the controller to precisely control the system state. The current sliding mode surface function defined in this paper is given by Equation 14:

$$\begin{cases} s_{id} = i_d^* - i_d \\ s_{iq} = i_q^* - i_q \end{cases} \tag{14}$$

Using a second-order sliding mode controller based on the super-twisting algorithm, precise control of the current loop in the Permanent Magnet Synchronous Motor (PMSM) can be achieved. Under this control strategy, the expressions of the current loop controller are given by Equations 15:

$$\begin{cases} u_d^* = K_{Pd} \left| s_{id} \right|^r \operatorname{sgn}(s_{iq}) + u_{sd} \\ \dfrac{d}{dt} u_{sd} = K_{Id} \operatorname{sgn}\left(s_{id} \right) \end{cases} \tag{15}$$

Let the design parameters be $r = 0.5$. the simulation block diagram of the current loop i_d controller based on SMC is shown in Figure 4, and the same is true for the current loop i_q design.

IV. RESULTS& DISCUSSION

In this paper, MATLAB/Simulink software is used to estimate the rotor position information of PMSM by STSMO, and the control strategy simulation experiments are carried out in combination with ADRC and SMC instead of speed loop and current loop respectively. The classical PMSM mathematical model under the d-q coordinate system is adopted in the system model, and the PMSM simulation model is shown in Figure 2.

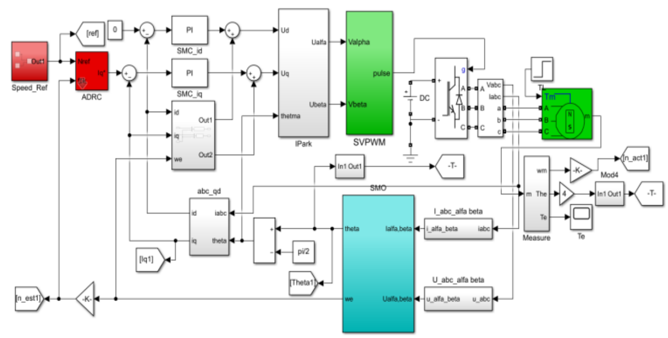

Figure 2: PMSM Simulation Model

A. Analysis of STSMO Simulation Results

In the simulation experiment, two observation methods, STSMO and SMO, were compared and analyzed. The comparison results are shown in Figure 3.

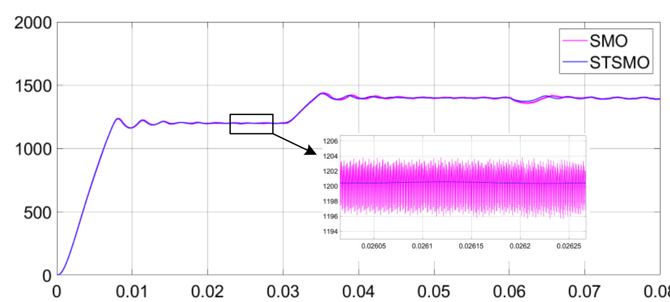

Figure 3: Comparison of Rotor Speed Estimation Between STSMO and SMO Observers

By comparing the above figure, it can be seen that the STSMO demonstrates superior performance compared to the SMO. The STSMO shows significant improvements in terms of system response speed, disturbance rejection, and long-term stability. Especially during the initial system startup and when the system is subjected to external disturbances, the STSMO can reach a steady state more quickly. The STSMO employs an advanced second-order sliding mode control strategy and a

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super-twisting algorithm, thereby endowing the system with stronger robustness and anti-interference capabilities. This design allows the system to maintain higher stability and control precision when facing various disturbances and uncertainty factors. In addition, the zoomed-in view further reveals the advantages of the STSMO in detail, especially in suppressing system fluctuations and maintaining stable operation.

B. Speed Analysis in Simulation Results of Control Strategy

In this section, to verify the advantages of the control strategy designed in this paper in the PMSM control system, both the improved ADRC-SMC and the traditional PI control strategy are used with the STSMO for predicting speed and angle, and a detailed comparative analysis is conducted to evaluate their performance in motor control. The speed comparison chart between the improved ADRC and the traditional PI control strategy is shown in Figure 4:

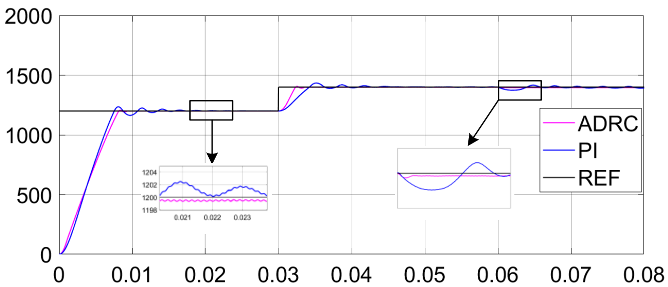

Figure 4: Speed Comparison Chart between ADRC_SMC and PI Control Strategies

According to the simulation results analysis, after the system reaches a steady state, the traditional PI control strategy produces a static error of about 1.3 r/min. The ADRC_SMC strategy designed in this paper significantly reduces the static error to only 0.5 r/min, and it can quickly recover to a steady state when the system encounters disturbances.

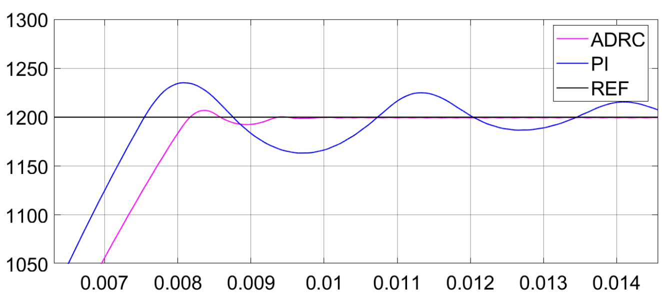

Figure 5: ADRC_SMC and PI control the local amplification of the rising curve speed

Through Figure 5, by comparing the local magnification analysis of the rising curve between ADRC_SMC and PI control, it can be determined that the overshoot of ADRC_SMC when the set speed changes is about 0.58%, while the overshoot of the PI control strategy when the set speed changes is about 2.92%. The results indicate that the ADRC_SMC strategy is significantly superior to the traditional PI control strategy in terms of control performance. Specifically, the overshoot of the ADRC_SMC control strategy is significantly less than that of the PI control strategy, and the duration of the overshoot is shorter.

By comparing the actual speed with the observed speed of the ADRC_SMC control strategy through the simulation diagram, as shown in Figure 6:

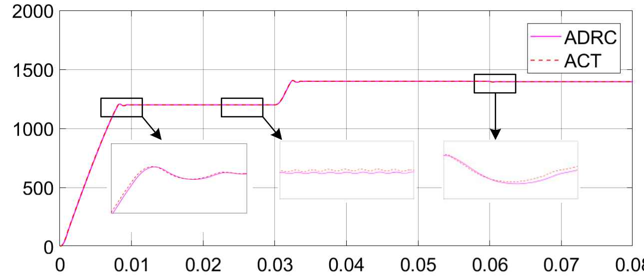

Figure6: ADRC_SMC Control Strategy's Actual Speed vs. Observed Speed Comparison Chart

Through the above chart, it can be seen that ADRC_SMC exhibits excellent tracking performance. The actual speed of the ADRC_SMC control strategy and the observed speed are almost completely superimposed, showing extremely high precision and stability.

C. Angle Analysis in Simulation Results of Control Strategy

The actual rotor position of the ADRC_SMC and the traditional proportional-integral PI control strategy is shown in Figure 7:

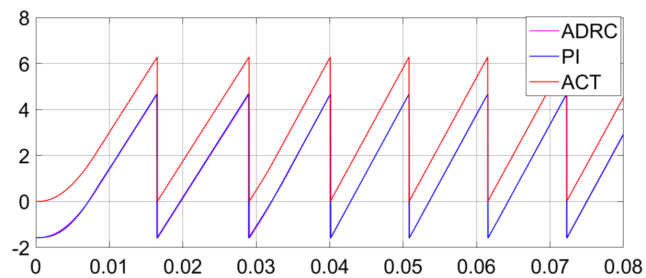

Figure 7: Comparison Chart of Actual Rotor Position between ADRC_SMC and PI Control

The local magnification figure is shown as Figure 8:

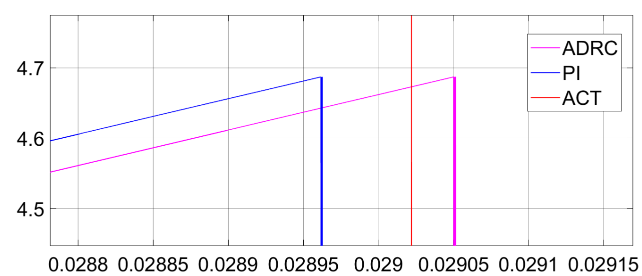

Figure 8: Local Magnification Chart of Actual Rotor Position Comparison between ADRC_SMC and PI Control

Through comparative analysis of the diagrams, significant differences can be observed between the ADRC_SMC and the traditional proportional-integral PI control strategy in terms of rotor angle tracking performance. The ADRC_SMC control strategy demonstrates higher precision and stability in rotor position estimation, with a high degree of consistency between the estimated rotor angle and the actual angle.

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V. CONCLUSIONS

In this paper, an improved control strategy that integrates Second-Order Super-Twisting Sliding Mode Observer (STSMO), Active Disturbance Rejection Controller (ADRC), and Sliding Mode Control (SMC) technologies is proposed to address the control issues of Permanent Magnet Synchronous Motors (PMSMs) under sensorless conditions. This strategy achieves high-precision control of PMSMs. The main results and conclusions are detailed as follows:

1. Application of Second-Order Super-Twisting Sliding Mode Observer (STSMO): This paper proposes the application of STSMO for the observation of the back electromotive force (EMF) of PMSMs. By adopting STSMO,the system maintains high-precision observation under conditions of step changes in speed or external disturbances.

2. Design and Optimization of ADRC and SMC Control Systems: This paper designs in detail an ADRC and SMC control system suitable for PMSMs, including the Tracking Differentiator (TD), Extended State Observer (ESO), and Nonlinear Error Feedback Control Law (NLESF). The optimization of system parameters ensures the stability and fast response capability of the control system.

3. Comparative Analysis with Traditional Control Methods: In simulation experiments, this paper compares traditional sensorless control methods for PMSMs with the improved method based on ADRC. The results show that the control system combining ADRC and SMC has faster response speed and better stability under speed step changes and disturbance conditions.

4. Simulation Verification of the Advantages of the Improved Control System: Through simulation experiments under various conditions, this paper demonstrates the significant advantages of the improved control system compared to traditional PI control. The improved control system exhibits smaller errors, faster response speed, stronger disturbance rejection capability, and higher control accuracy.

In summary, this paper successfully verifies the effectiveness of the Second-Order Super-Twisting Sliding Mode Observer, ADRC, and SMC control systems in sensorless control of PMSMs. The proposed method not only improves the response speed and observation accuracy of the system but also enhances its stability and disturbance rejection capability, providing a theoretical and practical foundation for practical applications. Future work will further optimize the control algorithm and conduct more experimental validations to promote the practical application of this method in industry.

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